MILITARY SPECIFICATION SHEET

RELAYS, HYBRID, SENSORS, ESTABLISHED RELIABILITY, TYPE I, CLASS B, (HYBRID AC POWER MONITORING SENSOR), 2PDT, LOW LEVEL TO 10 AMPERES, HERMETICALLY SEALED (POTTED), HORIZONTAL MOUNT WITH CONNECTOR TERMINATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-28894.

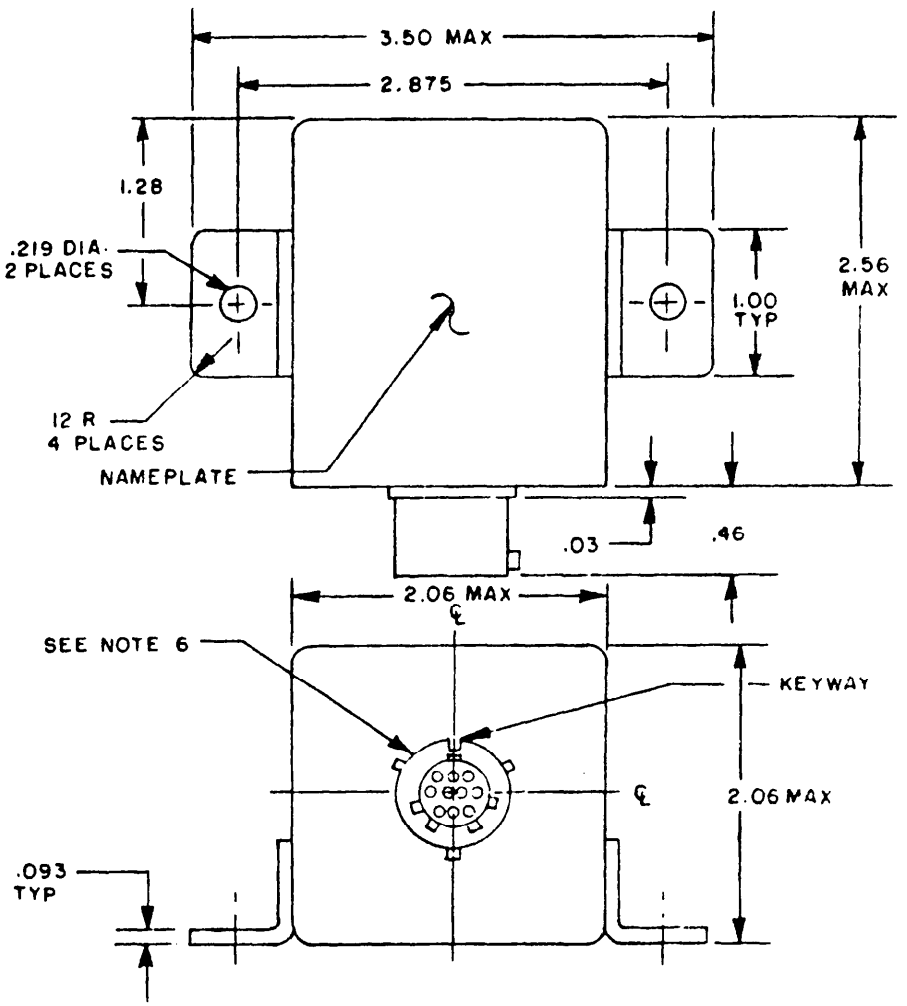
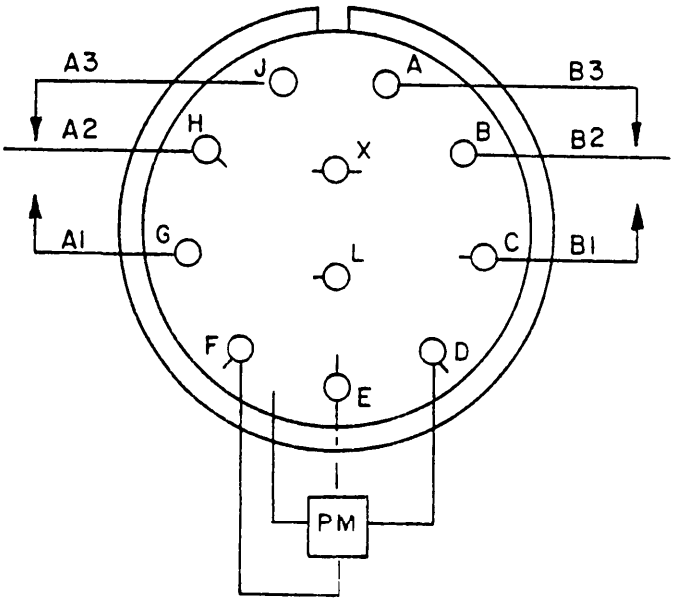



FIGURE 1. Outline dimensions and configuration.


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CIRCUIT DIAGRAM TERMINAL VIEW

| INCHES | MM |
|---|---|
| .03 | 0.8 |
| .093 | 2.36 |
| .219 | 5.56 |
| 1.00 | 25.4 |
| 1.28 | 32.5 |
| 2.06 | 52.3 |
| 2.56 | 65.0 |
| 2.375 | 73.03 |
| 3.50 | 88.9 |

| Pin | Input Connection |
|---|---|
| D | ØA |
| E | ØB |
| F | ØC |
| L | Neutral |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Terminal number shown above for reference only. Numbers do not appear on header.
4. Terminals 1, 2, 3, and 4 are non-functional leads.
5. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
6. Receptable connector, hermetic, boyonet coupling, solder-mounting per MS3113, part number 3113H1SC11P.

FIGURE 1. Dimensions and configuration - Continued.



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REQUIREMENTS:

Input data:

Input voltage range: 90 to 150 V rms, line to neutral, 3 phase.

Input frequency range: 44 to 450 Hz.

Sensing accuracy range (based on a true sinusoidal input waveform):

Voltage: Maximum ±2%.

Frequency: ±2%.

Operating mode:

a. The output relay will energize when all of the following conditions exist:

1. Each of the three phase voltages is within the under and over trip point limits.

2. The frequency is within the selected under and over trip point limits.

3. The phase rotation is ABC.

4. The pickup time delay period is completed.

b. The output relay will change to or remain in the de-energized state when any or all of the above conditions are not met (including an open circuit phase).

c. The output relay will change state (energize or de-energize) after the selected time delay, ±10%.

Operating data:

Pickup time (electromechanical relay): 50 milliseconds maximum when the requirements for operating mode are satisfied.

Dropout time (electromechanical relay): 50 milliseconds maximum when the requirements for operating mode are satisfied.

Timing action: Delay on operate, delay on release, or both.

Time delay: Fixed, select from .05 to 10 seconds.

Timing accuracy: ±10% of the nominal value.

Phase sequence sensing: ABC.

Maximum operating current per phase: 75 milliamperes rms.

Contact data:

Configuration: 2PDT, 10 amperes, integral class B relay.

Load ratings:

High level (relay case grounded):

Resistive:
10 amperes at 28 V dc, 115 V ac 400 Hz and 115/200 V ac 400 Hz.
2.5 amperes at 115 V ac 50/60 Hz and 115/200 V ac 50/60 Hz. 1/

Inductive:
8 amperes at 28 V dc, 115 V ac 400 Hz and 115/200 V ac 400 Hz (20,000 cycles minimum life).
2.5 amperes at 115 V ac 50/60 Hz and 115/200 V ac 50/60 Hz. 1/

1/ For 50/60 Hz, overload is not applicable and life shall be 10,000 cycles.



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Motor:
4 amperes at 28 V dc, 115 V ac 400 Hz and 115/200 V ac 400 Hz.
2 amperes at 115 V ac 50/60 Hz and 115/200 V ac 50/60 Hz. 1/

Lamp:
2 amperes at 28 V dc, 115 V ac 400 Hz, and 115/200 V ac 400 Hz.
1 ampere at 115 V ac 50/60 Hz and 115V/200 V ac 50/60 Hz. 1/

Low level: 10 to 50 microamperes at 10 to 50 millivolts (dc or peak ac).

Intermediate current: Applicable.

Contact voltage drop:

High level:

Before life: 0.150 volt maximum.

After life: 0.175 volt maximum.

Low level:

Before life: 0.05 ohm maximum.

After life: 0.15 ohm maximum.

Contact bounce: 1.0 millisecond maximum (applicable to failure rate level "L").

Contact stabilization time: 1.5 milliseconds maximum (applicable to failure rate levels "M", "P", and "R").

Overload: 40 amperes dc, 60 amperes ac.

Electrical data:

Insulation resistance: 100 megohms minimum initial; 50 megohms after environmental or life tests.

Dielectric withstanding voltage:

| | Initial | After life tests |
|---|---|---|
| Input to case: | 1000 V rms (60 Hz) | 1000 V rms (60 Hz) |
| All other points: | 1250 V rms (60 Hz) | 1000 V rms (60 Hz) |

| | Altitude | |
|---|---|---|
| | 80,000 ft | 300,000 ft |
| Input to case: | 350 V rms (60 Hz) | 500 V rms (60 Hz) |
| All other points: | 350 V rms (60 Hz) | 500 V rms (60 Hz) |

Voltage transients: MIL-STD-704, CAT-B.

Undervoltage trip point sensing range: 90 to 130 V rms.

Overvoltage trip point sensing range: 110 to 150 V rms.

Timing cycle: Applicable.

Environmental data:

Temperature range: -55°C to +125°C.

Thermal shock: Applicable.

1/ For 50/60 Hz, overload is not applicable and life shall be 10,000 cycles.



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Vibration: MIL-STD-202, method 204, test condition D, except frequency shall be 5 to 2,000 Hz.

Vibration, random: MIL-STD-202, method 214, test condition IG, 15 minutes each plane.

Shock: MIL-STD-202, method 213, test condition A (50G's).

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Physical data:

Seal: Hermetic, potted.

Dimensions and configuration: See figure 1.

Terminal strength: Method 211 of MIL-STD-202.

Pull test: Test condition A, 10 pound pull.

Twist test: Test condition D.

Solderability: Applicable.

Weight: 766 grams (27 ounces) maximum.

Identification marking (full): Applicable.

Life test requirements:

High level: 100,000 cycles minimum.

Low level: 100,000 cycles minimum plus 400,000 mechanical life.

Qualification inspection:

Qualification inspection and sample size: See table I.

TABLE I. Qualification inspection and sample size. 1/

| Single submission | Group submission | |
|---|---|---|
| 18 units plus 1 unsealed unit for level L at C = 0 2/ | M28894/4-AAAAA<br>M28894/4-MGGYY | 18 units plus 1 unsealed unit for level L at C = 0 2/ |
| 34 units plus 1 unsealed unit for level M at C = 0 2/ | | 34 units plus 1 unsealed unit for level M at C = 0 2/ |
| Qualification inspection as applicable. | | Qualification inspection as applicable. |
| | M28894/4-FBBKK<br>M28894/4-GCCLL<br>M28894/4-HDDMM<br>M28894/4-JEENN | 2 units each part number Qualification inspection, group 1. |

1/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-R-28894/1, MIL-R-28894/2, MIL-R-28894/3, MIL-R-288994/5, MIL-R-28894/6 and MIL-R-28894/7 maybe used in addition to MIL-R-28894/4 data.

2/ The number of units required for qualification testing will be increased as required in group V of the qualification inspection table of MIL-R-28894 if the contractor elects to test the number of units permitting one or more failures.



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Qualification inspection (reduced testing) and sample size: See table II. If the relays produced for MIL-R-28894/4 are similar in construction and design, except for the mounting means or the number of output contacts to the relays produced for MIL-R-28894/1, MIL-R-28994/2, MIL-R-28894/3, MIL-R-28894/5, MIL-R-28894/6, MIL-R-28894/7 then reduced testing for qualification of MIL-R-28894/4 relays may be performed concurrent with or subsequent to successful qualification of MIL-R-28894/1, MIL-R-28894/2, MIL-R-28894/3, MIL-R-28894/5, MIL-R-28894/6, MIL-R-28894/7.

TABLE II. Qualification inspection (reduced testing).

| Inspection |
|---|
| 4 units each dash number specified in table I<br>Group I of qualification inspection table plus shock and vibration<br>1 unsealed sample unit for internal examination |

Part number: Consists of the basic number of this specification sheet and a dash number denoting the frequency trip points, type of time delay and duration of time delay, undervoltage trip point, overvoltage trip point and failure rate level indicator.

Example: M28894/4- B C Z F N L

Basic specification number number.

Frequency trip points (see table III).

Duration of time delay (pickup) (see table IV).

Duration of time delay (dropout) (see table V).

Undervoltage trip point (see table VI).

Overvoltage trip point (see table VII).

Failure rate level (see table VIII).

TABLE III. Codes for frequency trip points.

| Code letter | Trip point (Hz) | |
|---|---|---|
| | Under frequency | Over frequency |
| A | 390 | 410 |
| B | 380 | 420 |
| C | 370 | 430 |
| D | 360 | 440 |
| E | 350 | 450 |
| F | 58 | 62 |
| G | 56 | 64 |
| H | 54 | 66 |
| J | 52 | 68 |
| K | 48 | 52 |
| L | 46 | 54 |
| M | 44 | 56 |



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TABLE IV. Codes for time delay on pickup.

| Code letter | Time delay on pickup (seconds) |
|---|---|
| A | .05 |
| B | .50 |
| C | 1 |
| D | 2 |
| E | 5 |
| F | 10 |
| G | .03 max |
| Z 1/ | Disabled |

1/ If timing condition is not required, specify "Z" to disable.

TABLE V. Codes for time delay on dropout.

| Code letter | Time delay on dropout (seconds) |
|---|---|
| A | .05 |
| B | .50 |
| C | 1 |
| D | 2 |
| E | 5 |
| F | 10 |
| G | .03 max |
| Z 1/ | Disabled |

1/ If timing condition is not required, specify "Z" to disable.

TABLE VI. Codes for undervoltage trip point. 1/

| Code letter | Sensing range 90 to 130 V rms | Code letter | Sensing range 90 to 130 V rms |
|---|---|---|---|
| A | 90 | M | 112 |
| B | 92 | N | 114 |
| C | 94 | P | 116 |
| D | 96 | R | 118 |
| E | 98 | T | 120 |
| F | 100 | U | 122 |
| G | 102 | V | 124 |
| H | 104 | W | 126 |
| J | 106 | X | 128 |
| K | 108 | Y | 130 |
| L | 110 | Z 2/ | Disabled |

1/ Specify trip point within indicated sening range using code letter.

2/ If undervoltage is not required, specify "Z" to disable.



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TABLE VII. Codes for overvoltage trip point. 1/

| Code letter | Sensing range 110 to 150 V rms | Code letter | Sensing range 110 to 150 V rms |
|---|---|---|---|
| A | 110 | M | 132 |
| B | 112 | N | 134 |
| C | 114 | P | 136 |
| D | 116 | R | 138 |
| E | 118 | T | 140 |
| F | 120 | U | 142 |
| G | 122 | V | 144 |
| H | 124 | W | 146 |
| J | 126 | X | 148 |
| K | 128 | Y | 150 |
| L | 130 | Z 2/ | Disabled |

1/ Specify trip point within indicated sening range using code letter.

2/ If undervoltage is not required, specify "Z" to disable.

TABLE VIII. Codes for failure rate level.

| Code letter | Failure rate level (% per 10,000 cycles) |
|---|---|
| L | 3.0 |
| M | 1.0 |
| P | 0.1 |
| R | 0.01 |

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AV
Navy - OS, SH

Preparing activity:
Navy - EC

Agent:
DLA - ES

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